

AMBIENT SYSTEMS

Climate At Your Service

www.ambientsystems.ai

Outline

1. Motivation
2. Problem and Market Trends
3. Solution and Technology
4. Applications
5.
6. Ambient Benefits
7. Appendix

Cyber-Physical Systems (CPS) are driving the next wave of Enterprise Applications

Machine Intelligence

Edge Devices

Cyber Security

Automation



Custom Applications

There is increasing demand for system optimization beyond energy efficiency

Problem

1. Limited Resources

It’s hard to find highly trained and experienced operators who can optimize multiple intelligent systems.

2. Technological Complexity

Many operate with outdated legacy systems and physical equipment that lack interoperability.

3. Competing Priorities

Different physical systems have different requirements based on the unique needs of occupants and operators.

Market Drivers

Productivity

1. Process Optimization
2. Enhanced Visibility
3. Faster Compliance

Sustainability

1. Energy Efficiency
2. Carbon Management

Cybersecurity

1. System Complexity
2. Increased Attack Surface

Applied AI for Cyber-Physical Systems





Sense

IoT gateway and sensors co-designed with manufacturing partners

Optimize

NUS National University of Singapore

Patent Pending Invention from National University of Singapore (NUS)

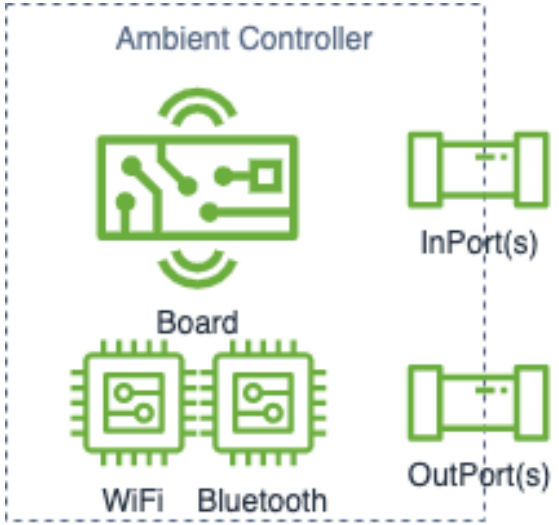


Control

Offering end-to-end equipment controls after development

Technology

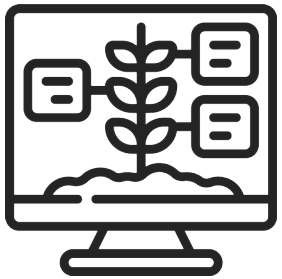
- IoT Gateway
  (Alternative Names: Hub, Terminal, Controller, Node)
  - Central hub for site-level equipment and networks
  - Communicates using various RFID frequencies and protocols
  - Ports for physical connection on site
  - Edge ML: device profiling and categorization
- User Application
  - Cloud-based web app (mobile to follow)
  - APIs for auxiliary systems
- Integrated Sensors
  - Integrated Air Sensors (Temperature, RH, CO2)
  - Additional Sensors Based on Use-Case



Software Stack

- Linux, C/C++
- Drivers: WiFi + Bluetooth
- Protocols: MQTT, ZigBee, LoRa, BacNET, Modbus
- Ports: Ethernet, RS-232

Applications



1. AgTech
Indoor farming sustainability & optimization



2. Healthcare
Patient well-being



3. Retail
Customer experience and inventory optimization



4. Consumer Products
Cyber security of smart home devices



5. Utilities
Smart grid operations and security

What Makes Ambient Different

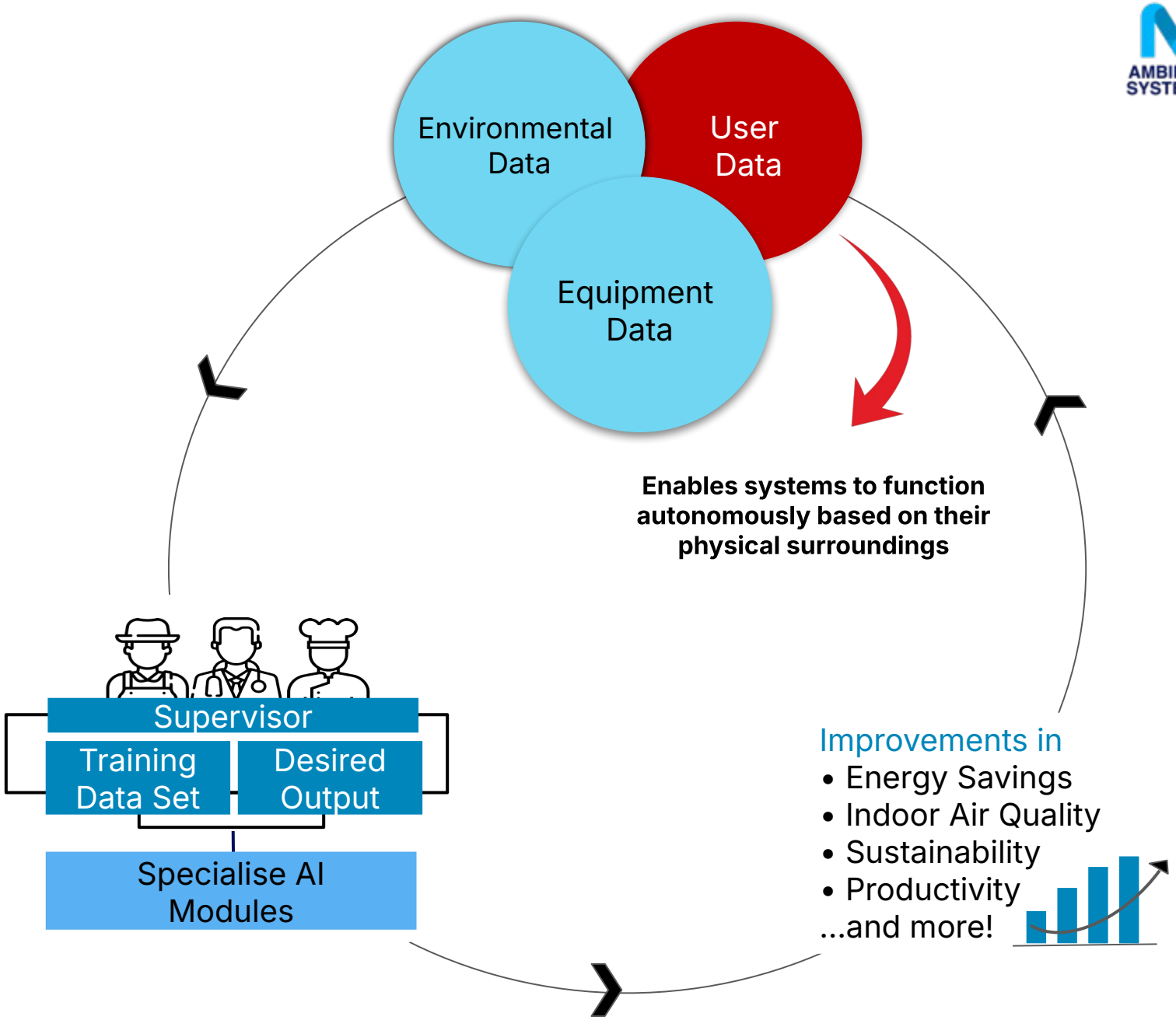
1. Our Methodology:
System optimization beyond energy efficiency.

2. Our IP:
Patent-pending invention for our launch market in AgTech.

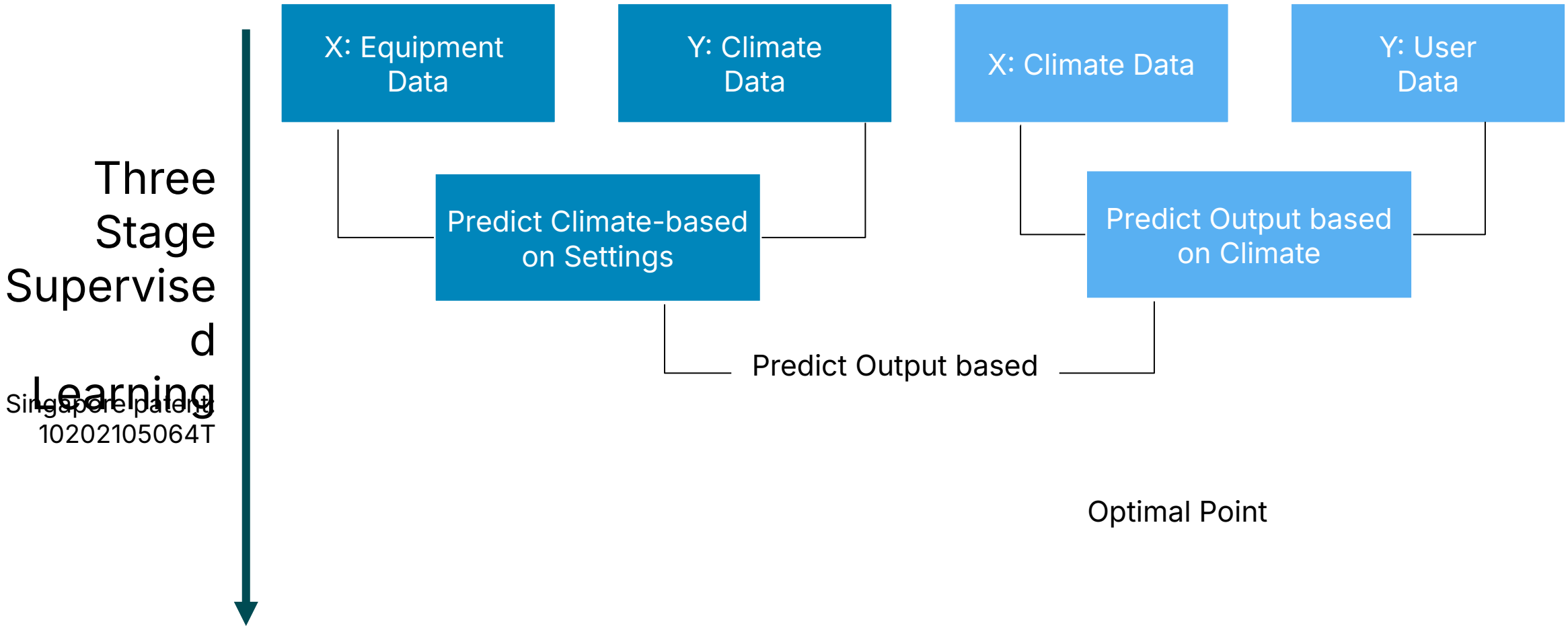
3. Our Trade-secret:
In-depth understanding of various industrial processes.

4. Our Traction:
University backing, 2 LOIs, pipeline of major companies.

System Optimization beyond Energy Savings



Underlying Magic: for Management of Ph

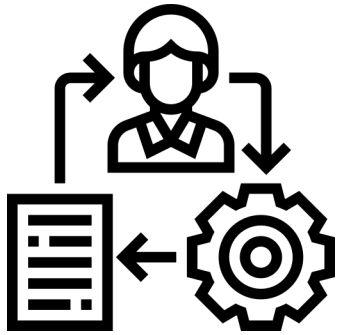


https://www.techinnovation.com.sg/website/2 optimization-for-indoor-farming/

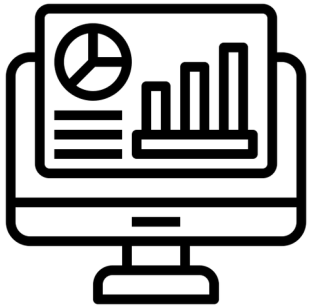
Feature #1: Productivity



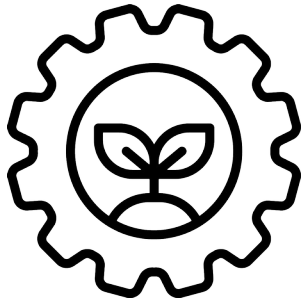
1. Evaluate the process needing improvement



2. Embed intelligent machines across the process

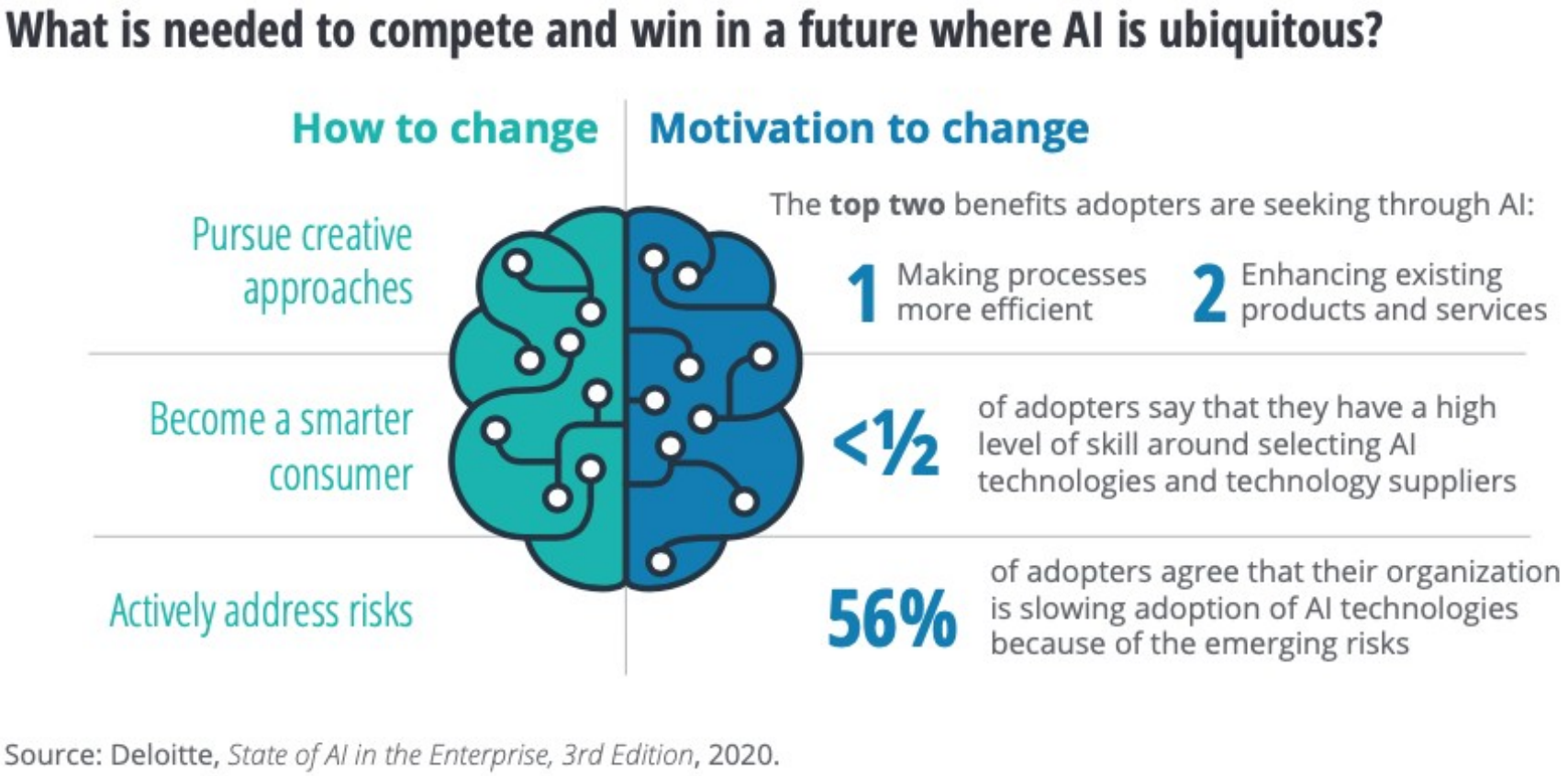


3. Leverage increased visibility, automated monitoring, compliance, and proprietary machine learning models from a single platform



4. Launch use-case: indoor farm sustainability and optimization.

Market Trends: AI-enabled Productivity



Source: Deloitte

Feature #2: Sustainability

1. Full visibility into certification requirements for sustainability.
2. Productivity needs balanced with energy efficiency.
3. Improved IAQ increased health and wellbeing of occupants, reducing healthcare costs and risks for respiratory illnesses.
4. Example: Indoor farms conditions and energy costs are optimized simultaneously.
5. Example: whole building ventilation is shifted to when electricity prices are lowest (e.g. night-time)



Increasing Focus on Sustainability

90%

of consumers are concerned about the Environment

Source: Sustainability Matters Now More Than Ever for Consumer Companies BCG, 2020

50%

of CXOs see tackling climate change as their generation's top priority

Source: Industry 4.0: At the intersection of readiness and responsibility Deloitte, 2020

Ambitious targets to green the built environment in Singapore by 2030:

80% green buildings

80% improvements in energy efficiency

80% new buildings to be super lower energy

Source: Industry 4.0: At the intersection of readiness and responsibility Deloitte, 2020

Source: SP Digital

Feature #3: Cyber Security

1. Site-level scripts use the device data stream to scan for anomalies.
2. Vulnerabilities are discovered and relayed to the user.
3. Leverages ML techniques for classification and anomaly detection.
4. Back-end Services delivered securely through AWS cloud.



Market Trends: Cyber Security



175zb

predicted collective sum of the world's data by 2025

Seagate 2018

65% of Organisations have a shortage of Cybersecurity staff



(ISC)² 2019

59% of Companies have experienced a data breach caused by a vendor or third party



Ponemon Institute 2018

51% of Cybersecurity professionals say their organisation is at risk due to Cybersecurity staff shortage



(ISC)² 2019

Source: Deloitte

Media and Links

- TechInnovation Tech Offer
- NUS GRIP Run 5 Post
- NUS GRIP Run 5 Media Post
- NUS GRIP Pitch
- LinkedIn: ambient-systems-ai
- Twitter: ambient_systems
- Crunchbase: ambient-systems
- Email: team@ambientsystems.ai

Thank You

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Appendix

Increasing Energy Demand and Consumption



Source: Our World in Data

In the built environment, it is imperative for buildings to be even more energy efficient and drive greater reduction in

Indoor Farming Overview

Improve crop yield in a market where only **51% of Indoor Farms are profitable**

24% reduction in loss of crops due to poor growing environment

Real-time improvements in farm operations efficiency resulting in **0.5 - 1.5 years** payback period



Indoor farm operators seeking to optimize crop performance through customized insights across key parameters.

Healthcare Overview

Prevent **65-70%** of hospital-associated infections costing up to USD11B

Hospitals with superior patient experience achieve **50%** more in net margins

Improve air filtration and recirculation on hospital premise to prevent nosocomial outbreaks of COVID-19 **(Mortality rates >60%)**



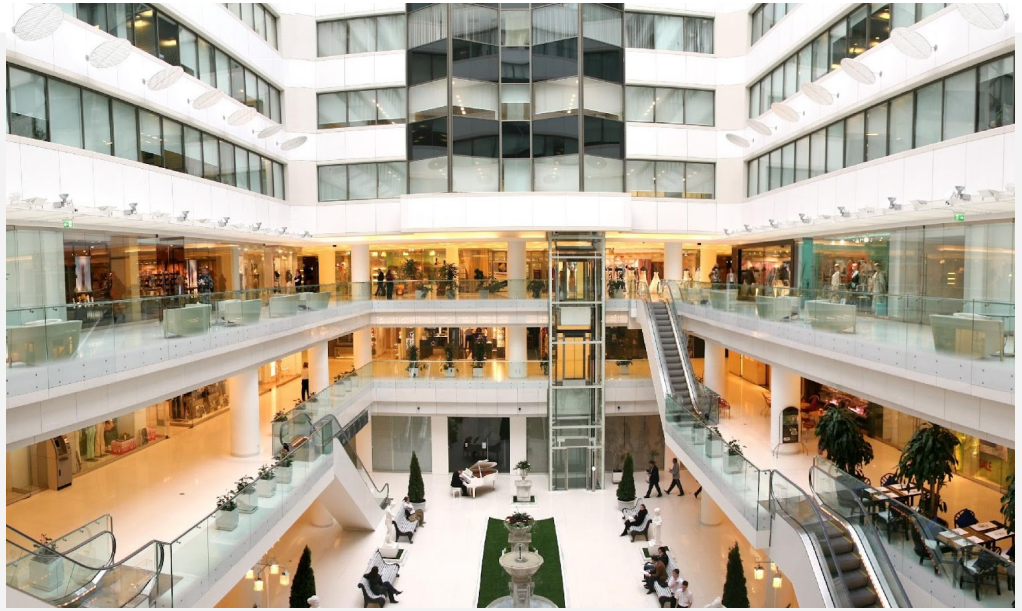
Hospitals looking to improve patient well-being hrough monitoring of indoor air quality

Retail Overview

70% of customers avoid retail stores if they are too hot, too cold or too stuffy

Using retail heatmaps to optimise retail layout resulting in **5%** increase in sales

Improve in-store conversions by more than **10%** by analysing customer footfall trends



Retailers looking to optimize customer experiences through improving retail comfort and analysing footfall behavior

Smart and Healthy Buildings Brings Value

Smart office buildings can potentially garner a **37%**

rent premium per nett square feet, compared to buildings without smart characteristics

Source: How Much Are Smart Buildings Really Worth? NAIOP Massachusetts, 2018

Healthy buildings increase productivity by **US$6,600 per employee**

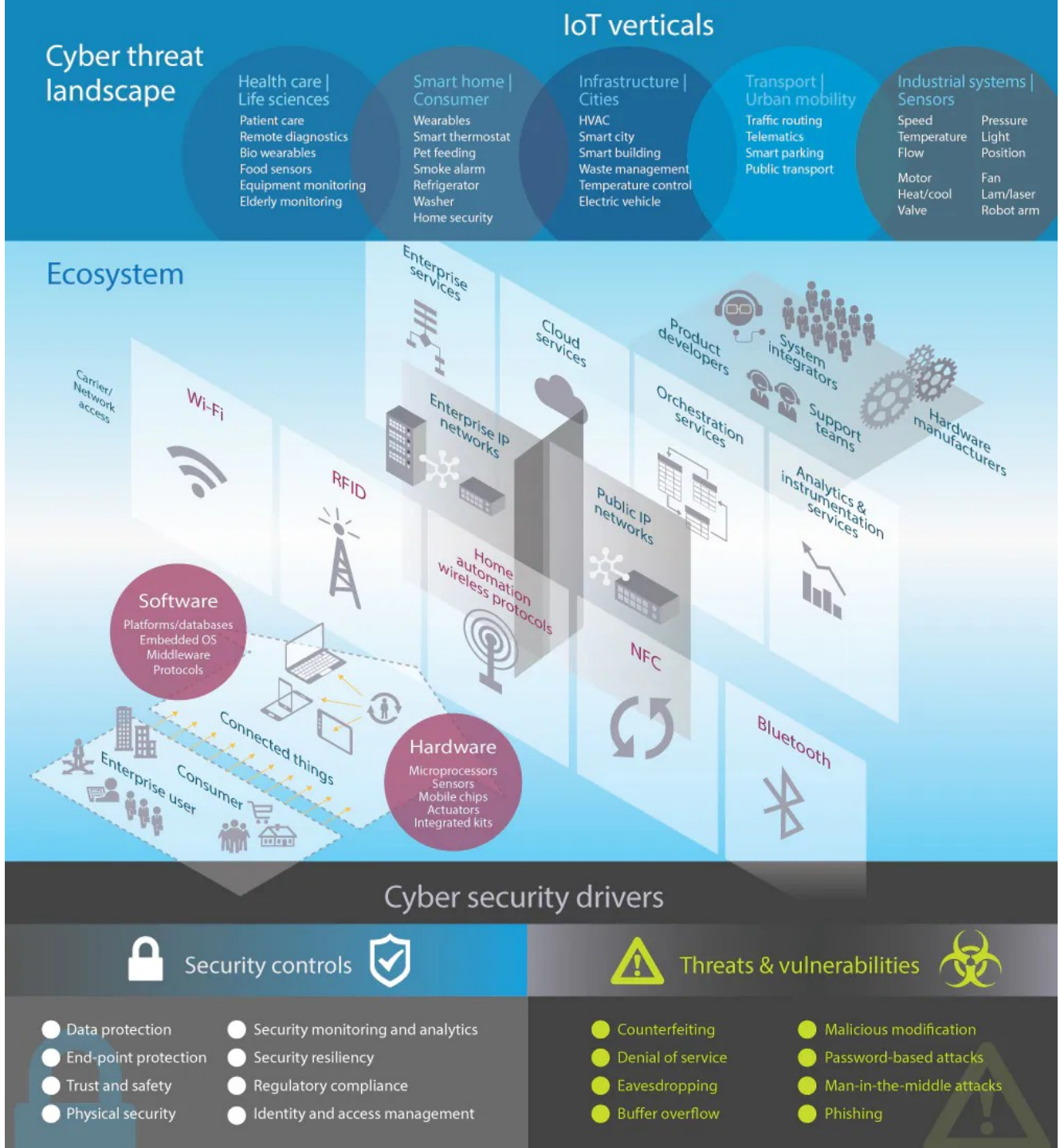
translating to cost savings and increased revenue for tenants

Source: The Best Talent will Demand the Smartest Buildings Memoori Smart Building Research, 2018

Building analytics can reliably drive **3-9%**

energy savings annually, a figure expected to persist or even increase

Source: Industry 4.0: At the intersection of readiness and responsibility Deloitte, 2020



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One step ahead with AI-enabled prescriptive maintenance

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Artificial Intelligence	Machine Learning	Deep Learning
		Data points gathered 
Automated Offerings • Building Performance Optimisation • Building Process Optimization	Smart Offerings • Asset Optimization • AI-based • AI-based Lighting Control	Cognitive Offerings • Predictive Maintenance

Identify general issues ✓

Understand energy consumption by analysing primary datapoints ✓

Personalised control ✓

Understand energy flow and building occupancy ✓

Collect external datapoints ✓